Backbone

Image 640 × 640

Prefix at b066

Conv × 3
32 → 64 → 64

Conv 128, s = 2

b009

SiLU

ELAN · 256

MP + Conv
Concat

b044

ELAN · 512

b066

P3 · 80²

MP + Conv, Concat
ELAN · 1024

P4 · 40²

MP + Conv, Concat
ELAN · 1024

P5 · 20²

Split b066: one boundary tensor

[1, 512, 80, 80] · 13.11 MB (FP32)

Blue frame: accelerator prefix
All subsequent modules: GPU suffix

b009 and b044: alternative cuts

Neck

Conv
128

Concat

Conv 128
Upsample ×2

ELAN · 256

Conv
256

Concat

Conv 256
Upsample ×2

SPPCSPC
512

ELAN · 128

MP + Conv

Concat

ELAN · 256

MP + Conv

Concat
ELAN · 512

Detection heads

RepConv 256
Conv 1×1 · 255

80 × 80
3 × 85

RepConv 512
Conv 1×1 · 255

40 × 40 · 3 × 85

RepConv 1024
Conv 1×1 · 255

20 × 20 · 3 × 85

COCO: 3 anchors × (4 box values + 1 objectness + 80 classes)